Because the Earth is so dynamic, rocks are always changing. Minerals that make up rocks are constantly moved and acted upon by the environment. Environmental agents change one type of rock into an entirely new rock. The web of environmental processes that forms and changes a rock is known as the rock cycle. You will learn about the three types of rocks: sedimentary, metamorphic, and igneous.

Write the three types of rocks you'll be learning about.
Check your spelling!

_____

fossils- remains or evidence of a living thing in rock

igneous rock- a rock formed from melted (molten) rock

metamorphic rock- a rock which is changed physically and even chemically when subjected to heat and pressure

sedimentary rock- crushed rock and organic material layered and compacted into new rock

extrusive rock- igneous rock formed from lava that solidifies on or near the earth's surface

intrusive rock- igneous rock formed when magma cools and solidifies beneath the earth's surface

Geologists, people who study rocks, believe that sedimentary rock was forming three and a half billion years ago.

Sedimentary rock is formed when mineral matter of plants and animals settle out of water and, less commonly, air or ice. The most common materials found in sedimentary rocks are fossils. Sedimentary rock covers 3/4 of the land area and most of the ocean floor.

Sedimentary Rocks

Read the information and trace the dotted words.

A g_____ is a person who studies rocks.

s_____ rock is formed when

mineral m_____of plants and animals settle out

of w_____.

The most common materials found in sedimentary rock are

f_____.

Sedimentary rocks cover 3/4 of the _____ area

and most of the _____floor.

Word Bank

geologist	**fossils**	**sedimentary**
matter	**land**	**water**
	ocean	

Sedimentary Rocks

Fill in the blanks and trace the dotted words.

A person who studies rocks is called a _____.

_____ rock is formed when

_____matter of plants and animals

settle out of water.

Sedimentary rock is believed to have started forming three and half

_____ years ago.

The most common materials found in sedimentary rock are

_____.

Sedimentary rock covers _____ of the land area and most of the ocean floor.

Word Bank

geologist	**billion**
sedimentary	**fossils**
mineral	**3/4**

Sedimentary Rocks

Fill in the blank with the correct word.

A person who studies rocks is called a _____.
geologist

_____ rock is formed when
Sedimentary

_____matter of plants and animals
mineral
settle out of water.

Sedimentary rock is believed to have started forming three and half

_____ years ago.
billion

The most common materials found in sedimentary rock are

_____.
fossils

Sedimentary rock covers _____ of the land area and most of the ocean floor.
3/4

Word Bank

geologist	**billion**
sedimentary	**fossils**
mineral	**3/4**

G	D	G	S	V	J	P	T	Q
M	E	F	N	X	V	L	A	G
I	R	O	C	K	A	A	N	D
N	E	S	L	N	X	N	I	V
E	T	S	D	O	M	T	M	X
R	T	I	S	C	G	S	A	D
A	A	L	O	E	P	I	L	V
L	M	S	Y	A	R	E	S	M
L	J	H	S	N	B	S	G	T

Word Bank

geologist	**fossils**	**ocean**
plants	**rock**	**mineral**
animals	**land**	**matter**

Sedimentary Rocks

Find the words in the Word Search.

Rock that has been altered by heat or by heat and pressure is called metamorphic rock. The word 'metamorphic' means 'change of form.' The types of minerals they contain and their textures identify metamorphic rocks. Mineral content depends on the conditions the rocks endured. Sedimentary rocks are changed to metamorphic rock when heat and pressure are applied. For example, limestone changing to marble, and shale changing to slate.

Remember: (Write the answers by using the information above.)

What kind of rock means "change of form"?

What kind of rocks are changed to matamorphic rock when heat and pressure are applied?

M_____ rocks are rocks that have been altered by heat and pressure.

The word 'metamorphic' means 'C_____ of form.'

Metamorphic rocks are identified by the types of m_____ they contain and their t_____.

M_____ content depends on the conditions the rock endured.

Sedimentary rocks are changed to metamorphic rocks when h_____ and p_____ are applied.

Word Bank

metamorphic	**textures**	**pressure**
change	**mineral**	
minerals	**heat**	

Metamorphic Rocks

Fill in the blanks.

1. What has altered metamorphic rock?

heat and pressure ice and lava water

2. What does 'metamorphic' mean?

stay the same change of form sits still

3. How are metamorphic rocks identified?

their color their shape types of minerals

4. What does the mineral content depend on?

conditions the rock endured location insects

5. Sedimentary rocks are changed into metamorphic rocks when what is applied?

water and salt heat and pressure rain and ice

Metamorphic Rocks

Circle the correct answer.

1. What has altered metamorphic rock?

heat and pressure ice and lava water

2. What does 'metamorphic' mean?

stay the same change of form sits still

3. How are metamorphic rocks identified?

their color their shape types of minerals

4. What does the mineral content depend on?

conditions the rock endured location insects

5. Sedimentary rocks are changed into metamorphic rocks when what is applied?

water and salt heat and pressure rain and ice

Example:
R obert likes to swim in the
O cean because he can see
C reatures and rocks that
K eep the ocean interesting.

M _____

E _____

T _____

A _____

M _____

O _____

R _____

P _____

H

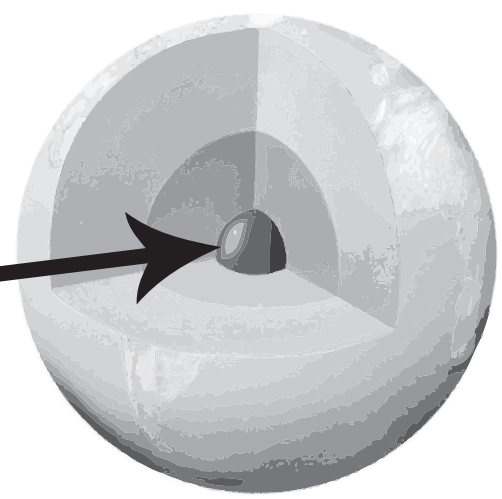
I

C

Metamorphic Rocks

Make an acrostic poem with the word 'metamorphic.'

In Northern Canada, much of the hard rock found is igneous rock. Igneous rock is formed by the hardening and crystallization of molten material that originates deep within the earth. 'Igneous' comes from the Greek word for 'fire'. Rocks that have hardened from magma, the hot liquid beneath the earth's crust, are igneous rocks. The properties and sizes of various crystals depend on the magma's composition and its rate of cooling. There are two types of igneous rocks: intrusive and extrusive.



What is the word for molten rock?

What kind of rock is named after the greek word for "fire"?

Igneous Rocks

Trace the dotted words, write and check your answers.

1. Most of the hard rock found in Northern

C_____ is called I_____ rock.

2. Igneous rock is rock formed by the hardening and crystallization

of M_____ material that originates deep within

the E_____.

3. 'Igneous' comes from the Greek word for 'F_____.'

4. Igneous rocks are rocks that have hardened from

M_____, the hot L_____ beneath

the earth's crust.

5. The P_____ and

S_____ of the various crystals depend on the

magma's composition and its rate of

C_____.

cooling	**size**	**earth**	**properties**
liquid	**magma**	**Canada**	
fire	**molten**	**igneous**	

Igneous Rocks

Fill in the blanks.

- where most igneous rock is found
- 'igneous' means
- hot liquid beneath the earth
- igneous rock is formed by the hardening and crystallization of
- properties and sizes of variouscrystals depend on
- molten means
- The center if the earth is called its

- magma's composition
- melted
- molten material
- core
- fire
- magma
- Northern Canada

Igneous Rocks

Draw an arrow to the correct word.

where most igneous rock is found

'igneous' means

hot liquid beneath the earth

igneous rock is formed by the hardening and crystallization of

properties and sizes of variouscrystals depend on

molten means

The center if the earth is called its

magma's composition

melted

molten material

core

fire

magma

Northern Canada

sneioug _____

kcro _____

olentm _____

aammg _____

refi _____

rathe _____

Igneous Rocks

Unscramble the words.

Igneous rocks are divided into two groups: extrusive and intrusive. Extrusive are formed when magma flows onto the surface of the earth or the floor of the ocean through deep cracks or fissures and at volcanic vents. Then it cools and hardens. Extrusive rocks have a fine-grained texture.

Igneous Rocks

Trace the dotted words.

1. _____ rocks are divided into _____ groups: _____ and intrusive.

2. Extrusive rocks form when _____ flows onto the surface of the _____ or floor of the _____ through _____ cracks or fissures and at _____ vents.

3. _____ rocks have a fine-_____ texture.

Word Bank

extrusive	**earth**	**volcanic**
grained	**ocean**	**igneous**
magma	**deep**	**two**

Fill in the blanks and trace the dotted words.

1. Igneous rocks are divided into 2 groups: _____ and intrusive.

a. sedimentary b. metamorphic c. extrusive

2. Extrusive rocks form when _____ flows onto the surface of the earth.

a. magma b. lava c. water

3. After the magma flows onto the earth's surface it then _____ and hardens.

a. boils b. cools c. does nothing

4. Extrusive rocks have a fine, _____ texture.

a. smooth b. colorful c. grained

5. Magma flows to the Earth's surface through _____.

a. cracks & volcanic vents b. mountains c. rivers

1. Igneous rocks are divided into 2 groups: _____ and intrusive.

a. sedimentary b. metamorphic c. extrusive

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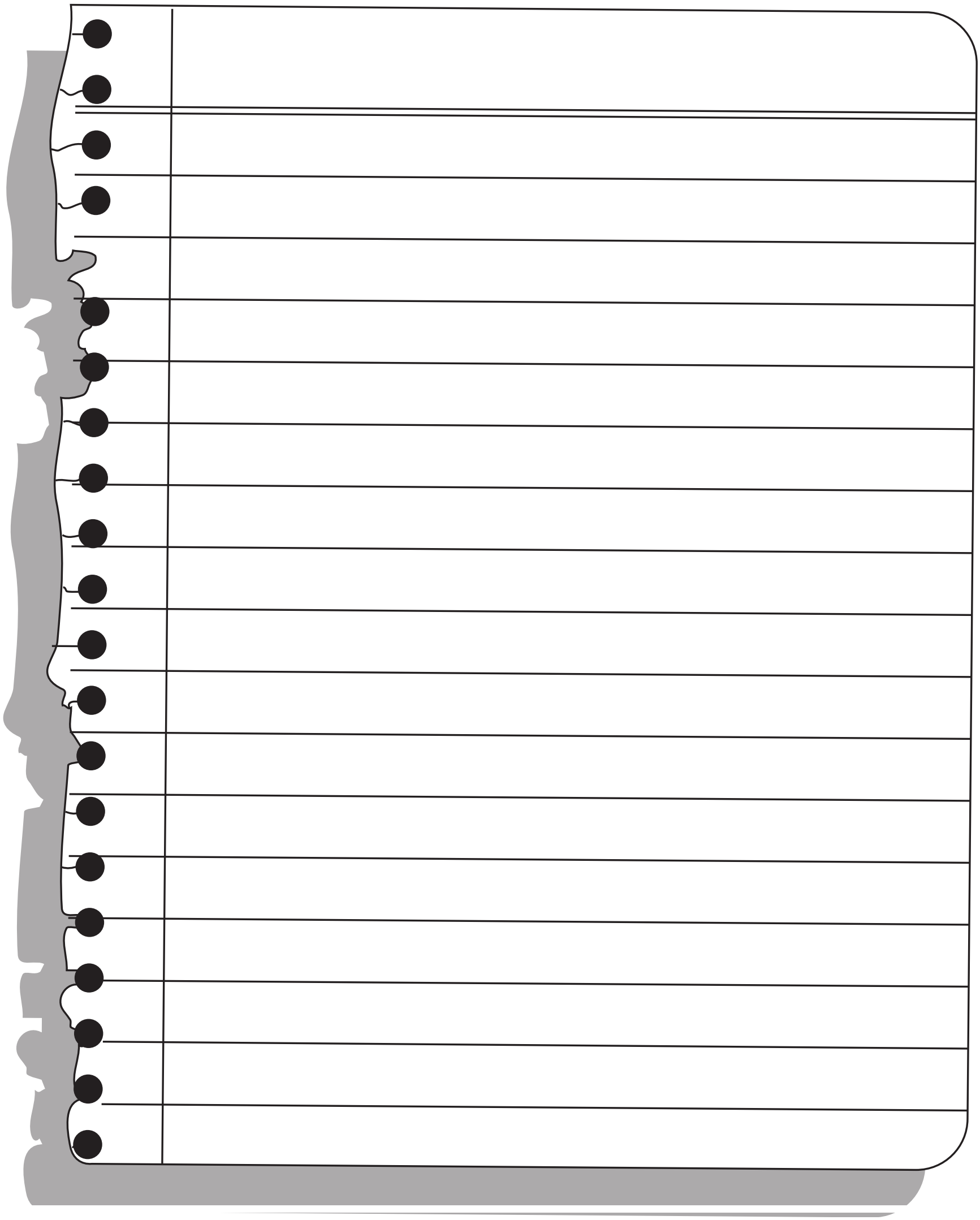
a. smooth b. colorful c. grained

5. Magma flows to the Earth's surface through _____.

a. cracks & volcanic vents b. mountains c. rivers

Example: tie, stick...



Igneous Rocks

How many words can you make from any of the letters in "Extrusive Rocks?"

Intrusive rocks are the other type of igneous rock. Intrusive rocks are made when magma solidifies beneath the earth's surface in mines and tunnels or at the surface where it has been exposed by geographical uplifting and by erosion. Intrusive rocks vary from thin sheets to huge irregular masses. Since magma that forms intrusive rocks solidifies slowly, most intrusive rocks have larger crystals than extrusive rocks.

The other type of igneous rock is _____.

What type of uplift exposes intrusive rocks?

_____

Intrusive rocks form from magma that cools slowly having larger _____.

I_____ rocks are made when

m_____ solidifies beneath the earth's

s_____ in mines and tunnels or at the surface

when it has been exposed by g_____

uplifting and by e_____.

The s_____ of intrusive rocks vary from

t_____ sheets to h_____ irregular masses.

Since magma that forms intrusive rocks

s_____ slowly, most intrusive rocks have

larger c_____ than

e_____ rocks.

Word Bank

extrusive	**intrusive**	**size**
solidifies	**magma**	**thin**
erosion	**crystals**	**geographical**
surface	**huge**	

Intrusive Rocks

Fill in the blanks using the word bank.

_____ made when magma solidifies beneath the earth's surface.

_____ intrusive rocks belong to this group

_____ intrusive rocks have been exposed by geographical uplifting and what else?

_____ intrusive rocks vary in what?

_____ intrusive rocks have bigger crystals than what other rock?

_____ magma that forms intrusive rocks solidifies how?

a. size

b. slowly

c. intrusive rocks

d. erosion

e. extrusive

f. igneous rocks

Intrusive Rocks

Write the letter for the correct answer in each blank.

_____ made when magma solidifies beneath
c
the earth's surface.

_____ intrusive rocks belong to this group
f

_____ intrusive rocks have been exposed by
d
geographical uplifting and what else?

_____ intrusive rocks vary in what?
a

_____ intrusive rocks have bigger crystals
e
than what other rock?

_____ magma that forms intrusive rocks
b
solidifies how?

a. size

b. slowly

c. intrusive rocks

d. erosion

e. extrusive

f. igneous rocks

Create your own rock band! Draw a picture of your rock band. It can have as many people and instruments as you like. When you are finished, give your rock band a name.
Examples: The Mad Magmas or The Mighty Igneous Fossils

So how does the rock cycle work? Its starts as rocks are pushed up by tectonic forces and eroded by wind and rain. The eroded rocks travel by wind or moving water until they are deposited, settling into layers. More eroded rocks may squeeze and press the layers into sedimentary rocks. Rocks can also be sunk down into lower layers of the earth by plate tectonic processes. Buried rocks may also melt and recrystallize into igneous rocks. Metamorphic, sedimentary, and igneous rocks may then be pushed up by tectonic forces, starting the rock cycle again.

The three kinds of rocks are?

_____ _____

_____

Check your spelling!

How it ROCKS! *Trace the dotted words.*

R_____ are pushed up by tectonic forces and

E_____ by wind and rain.

The eroded rocks travel by W_____ or moving

W_____ until they are deposited, settling into

L_____.

Sedimentary rocks are formed when eroded rocks

S_____ together and press the layers.

B_____ rocks may also M_____ and

recrystallize into igneous rocks.

M_____,

S_____, and I_____

rocks may then be pushed up and start the rock C_____

again.

rocks	**layers**	**metamorphic**
eroded	**squeeze**	**sedimentary**
wind	**buried**	**igneous**
water	**melt**	**cycle**

How it ROCKS!

Fill in the blanks using the Word Bank.

_____ More eroded rocks may squeeze and press the layers into sedimentary rocks.

_____ Buried rocks may also melt and recrystallize into igneous rocks.

_____The eroded rocks travel by wind or moving water until they are deposited, settling into layers.

_____ Metamorphic, sedimentary, and igneous rocks may then be pushed up by tectonic forces, starting the rock cycle again.

_____ Rocks are pushed up by tectonic forces and eroded by the wind and rain.

_____ More eroded rocks may squeeze and press the layers into
3

sedimentary rocks.

_____ Buried rocks may also melt and recrystallize into igneous rocks.
4

_____The eroded rocks travel by wind or moving water until they are
2

deposited, settling into layers.

_____ Metamorphic, sedimentary, and igneous rocks may then be
5

pushed up by tectonic forces, starting the rock cycle again.

_____ Rocks are pushed up by tectonic forces and eroded by the
1

wind and rain.

S	I	K	N	B	B	H	L	V	C	X	M
X	E	A	R	T	H	C	R	V	B	N	E
N	R	D	M	L	K	J	A	H	G	F	T
F	O	W	I	N	D	D	I	D	S	A	A
Q	S	W	E	M	R	T	N	S	Y	U	M
I	I	O	P	A	E	M	N	R	B	V	O
V	O	C	X	G	Z	N	S	E	K	D	R
F	N	J	K	M	L	O	T	Y	C	I	P
I	U	Y	T	A	R	F	G	A	O	H	H
J	K	G	F	H	B	N	M	L	R	O	I
I	G	N	E	O	U	S	U	Y	T	Y	C

Word Bank

rock	**layers**	**igneous**
erosion	**earth**	**magma**
wind	**sedimentary**	
rain	**metamorphic**	

How it ROCKS!

Find the words in the Word Search.

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